# Zander W. Blasingame

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## Professional Experience

**AITHYRA** *Vienna, Austria* — **Postdoctoral Fellow** — 2026 – Present
*Supervisor: Alexander Tong*
• Stochastic flow maps, likelihoods for continuous normalizing flows, AI for Science

**Clarkson University** *Potsdam, NY* — **Graduate Research Assistant** — 2018 – 2025
*Supervisor: Chen Liu*
• Diffusion models, neural differential equations, flow-based generative modeling
• Face morphing attacks, biometrics, anomaly detection

**Clarkson University** *Potsdam, NY* — **Undergraduate Research Assistant** — 2016 – 2018
*Supervisor: Chen Liu*
• GAN inversion, probability density estimation, zero-day malware detection

**Griffiss Institute** *Rome, NY* — **Engineering Intern** — 05.2016 – 08.2016

**UNH InterOperability Lab** *Durham, NH* — **Engineering Intern** — 07.2014 – 08.2014

## Education

**Clarkson University** *Potsdam, NY*

Ph.D. in Electrical and Computer Engineering — 2025
*Thesis: On Guided and Reversible Solvers for Neural Differential Equations*
M.Sc. in Electrical and Computer Engineering (en route to Ph.D.) — 2025
B.Sc. in Computer Engineering, *with Great Distinction* — 2018

## Publications

In reverse chronological order   *Denotes equal contribution   †Denotes equal advising

**ICML** *Oral* — **Rex: A Family of Reversible Exponential (Stochastic) Runge–Kutta Solvers** — 2026
***Z.W. Blasingame*** *and C. Liu*
Published at ICML (2026) · *Oral [top 0.7% of submissions]*
*Presented at ICLR 2026 DeLTa and ReALM-GEN workshops*
arxiv openreview code website pdf poster slides

**ICML** *Workshop* — **Strong Stochastic Flow Maps** — 2026
*S. McCallum*, **Z.W. Blasingame***, T. Herschell, N. Rindtorff, A. Tong†, and J. Foster†*
*Presented at ICML 2026 SPIGM workshop*
arxiv openreview code pdf slides video

**arXiv** — **LoRA as a Flexible Framework for Securing Large Vision Systems** — 2026
*R.E. Neddo, E.A. Atindama,* ***Z.W. Blasingame***†*, and C. Liu†*
Preprint (2026)
arxiv pdf

**ICLR** *Workshop*

**Stochastic Few-step Models** 2026
*R. Passaro, **Z.W. Blasingame**, M.M. Bronstein, and A. Tong*
*Presented at ICLR 2026 DeLTa and ReALM-GEN workshops*
openreview pdf

**NeurIPS**

**Greed is Good: A Unifying Perspective on Guided Generation** 2025
***Z.W. Blasingame** and C. Liu*
Published at NeurIPS (2025)
*Presented at ICML 2025 EXAIT workshop and ICLR 2025 FPI workshop*
arxiv openreview pdf poster slides

**IEEE MOST**

**Poster: Adapting Pretrained Vision Transformers with LoRA Against Attack Vectors** 2025
*R.E. Neddo, S. Willis, **Z.W. Blasingame**, and C. Liu*
Published at IEEE MOST (2025)

**IEEE ASAP**

**Scalable Malware Detection Framework Using Performance Counters and Gradient Boosting** 2025
*C. Woralert, C. Liu, and **Z. Blasingame***
Published at IEEE ASAP (2025)

**HASP**

**Feasibility of Scalable Performance Counter Malware Detection Framework Using LightGBM** 2025
*C. Woralert, C. Liu, and **Z. Blasingame***
Published at HASP (2025)

**ICLR** *Workshop*

**A Reversible Solver for Diffusion SDEs** 2025
***Z.W. Blasingame** and C. Liu*
*Presented at ICLR 2025 DeLTa workshop*
arxiv openreview pdf poster

**NeurIPS**

**AdjointDEIS: Efficient Gradients for Diffusion Models** 2024
***Z.W. Blasingame** and C. Liu*
Published at NeurIPS (2024)
*Presented at NeurIPS - New Frontiers in Adversarial Machine Learning*
arxiv openreview code website pdf poster slides

**IEEE S&P**

**Fast-DiM: Towards Fast Diffusion Morphs** 2024
***Z.W. Blasingame** and C. Liu*
Published at IEEE S&P (2024)
arxiv code website pdf

**IJCB** *Spotlight*

**Greedy-DiM: Greedy Algorithms for Unreasonably Effective Face Morphs** 2024
***Z.W. Blasingame** and C. Liu*
Published at IJCB (2024) · *Spotlight*
arxiv code website pdf poster

**IEEE TBIOM**

**Leveraging Diffusion for Strong and High Quality Face Morphing Attacks** 2024
***Z.W. Blasingame** and C. Liu*
Published at IEEE TBIOM (2024)
*Presented at IJCB (Oral)*
arxiv code website pdf poster slides

**IJCB** | **The Impact of Print-Scanning in Heterogeneous Morph Evaluation Scenarios** | 2024
*R.E. Neddo, **Z.W. Blasingame**, and C. Liu*
Published at IJCB (2024)
arxiv pdf poster

**HASP** | **Towards Effective Machine Learning Models for Ransomware Detection via Low-Level Hardware Information** | 2024
*C. Woralert, C. Liu, and **Z. Blasingame***
Published at HASP (2024)

**IEEE TCAS-I** | **HARD-Lite: A Lightweight Hardware Anomaly Realtime Detection Framework Targeting Ransomware** | 2023
*C. Woralert, C. Liu, and **Z. Blasingame***
Published at IEEE TCAS-I (2023)

**AsianHOST**
*Oral*
**A Comparison of One-class and Two-class Models for Ransomware Detection via Low-level Hardware Information** | 2023
*C. Woralert, C. Liu, **Z. Blasingame**, and Z. Yang*
Published at AsianHOST (2023) · *Oral*

**AsianHOST**
*Best Paper Nominee*
**HARD-Lite: A Lightweight Hardware Anomaly Realtime Detection Framework Targeting Ransomware** | 2022
*C. Woralert, C. Liu, and **Z. Blasingame***
Published at AsianHOST (2022) · *Best Paper Nominee*

**IJCB**
*Oral*
**Leveraging Adversarial Learning for the Detection of Morphing Attacks** | 2021
***Z. Blasingame** and C. Liu*
Published at IJCB (2021) · *Oral*
slides

**ICANN**
*Oral*
**Feature Creation Towards the Detection of Non-control-Flow Hijacking Attacks** | 2021
***Z. Blasingame**, C. Liu, and X. Yao*
Published at ICANN (2021) · *Oral*
slides

**HASP** | **Detecting Non-Control-Flow Hijacking Attacks Using Contextual Execution Information** | 2019
*G. Torres, Z. Yang, **Z. Blasingame**, J. Bruska, and C. Liu*
Published at HASP (2019)

**RESEC** | **Detecting Data Exploits Using Low-Level Hardware Information: A Short Time Series Approach** | 2018
*C. Liu, Z. Yang, **Z. Blasingame**, G. Torres, and J. Bruska*
Published at RESEC (2018)

**SPIE** | **Verification of OpenSSL Version via Hardware Performance Counters** | 2017
*J. Bruska, **Z. Blasingame**, and C. Liu*
Published at SPIE (2017)

## Invited Talks

**MolSS**
*Online*
**Rex: A Family of Reversible Exponential (Stochastic) Runge-Kutta Solvers** | 08.2026
*Machine Learning for Molecule Simulations & Sampler Reading Group*
slides

**AITHYRA** — *Vienna, Austria*
**Strong Stochastic Flow Maps** — 07.2026
*AI x Bio Seminar*
slides

**MolSS** — *Online*
**Strong Stochastic Flow Maps** — 07.2026
*Machine Learning for Molecule Simulations & Sampler Reading Group*
slides

**Starkly Speaking** — *Online*
**Strong Stochastic Flow Maps** — 07.2026
*Starkly Speaking Reading Group. Organized by Hannes Stärk*
slides video

**Mila** — *Montréal, Canada*
**AdjointDEIS: Efficient Gradients for Diffusion Models** — 01.2025
*Sampling Reading Group. Organzied by Kirill Neklyudov*
slides

**Embassy of the Kingdom of the Netherlands** — *Washington, D.C.*
**Diffusion Morphs (DiM): Diffusion is all you need for highly effective face morphs** — 11.2024
*Transatlantic Dialogue on Presentation Attack Detection*
slides

**IEEE IJCB** — *Buffalo, NY*
**Diffusion Morphs (DiM): Leveraging Diffusion for Strong and High-Quality Face Morphing Attacks** — 09.2024
*Journal Presentation*
slides

**Idiap** — *Martigny, Switzerland*
**Diffusion Morphs (DiM): The Power of Iterative Generative Models for Attacking FR Systems** — 07.2024
*Biometrics Security & Privacy Group*
slides

**DSA** — *Online*
**Diffusion for the Generation of Face Morphs** — 02.2024
*CITeR & Document Security Alliance*
slides

## Awards and Honors

ICML Oral Presentation [top 0.7% of submissions] — 2026

ICML Gold Reviewer Award — 2026

NeurIPS Top Reviewer Award — 2025

AISTATS Best Reviewer Award — 2025

IJCB Doctoral Consortium — 2024

Best Oral Presentation for Software Engineering — Clarkson RAPS — 2017

Clarkson Presidential Scholar — 2015 – 2018

## Teaching

**Griffiss Institute** — *Rome, NY*
**Instructor** — 2020 – 2024

- Cyber Summer Camp
- Advanced Cyber Summer Camp

**Clarkson University** *Potsdam, NY* — **Graduate Teaching Assistant** — 2018 – 2021

- Electrical Science
- Digital Design
- ECE Sophomore Lab

**Clarkson University** *Potsdam, NY* — **Undergraduate Teaching Assistant** — 2016 – 2018

- Software System Architecture
- Differential Equations

## Funding

**CITeR** — **Explainable Image Quality with Transformer-based Models** — 2023 – 2024
*NSF Industry–University Cooperative Research Center. Primary student investigator. PI: Chen Liu*

**CITeR** — **Towards the Creation of a Large Dataset of High-Quality Face Morphs** — 2021 – 2024
*NSF IUCRC. Primary student investigator at Clarkson. PIs: Chen Liu, Stephanie Schuckers, Xin Li, Jeremy Dawson, Nasser Nasrabadi, David Doermann, Srirangaraj Setlur, Siwei Lyu, Xiaoming Liu, Sébastian Marcel*

**CITeR** — **Comparative Detection of Facial Image Manipulation Techniques** — 2020 – 2022
*NSF IUCRC. Primary student investigator. PI: Chen Liu*

**CITeR** — **Adversarial Learning Based Approach Against Face Morphing Attacks** — 2019 – 2021
*NSF IUCRC. Primary student investigator. PI: Chen Liu*

## Professional Services

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|---|---|
| **Organizer** | IJCB Workshop on Face Morphing Attack and Detection Techniques (2024) |
| **Reviewer at NeurIPS** | 2024, 2025 (Top Reviewer), 2026 |
| **Reviewer at ICML** | 2025, 2026 (Gold Reviewer) |
| **Reviewer at ICLR** | 2026 |
| **Reviewer at AISTATS** | 2025 (Best Reviewer) |
| **Workshop Reviewer** | NeurIPS Frontiers in Probabilistic Inference (2025); ICCV 2COOOL (2025); ICLR Deep Generative Models (2025); ICLR Frontiers in Probabilistic Inference (2025); IJCB Face Morphing Attack and Detection Techniques (2024) |
| **Other Reviewing** | IEEE ICPR (2024); IEEE MOST (2023–2024); PMAM (2022–2023); ICANN (2021); IEEE BTAS (2019) |

## Skills

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|---|---|
| **ML Architectures** | Diffusion Models, Flow Matching, Neural Differential Equations, GANs, VAEs |
| **Distributed Computing** | Slurm, CUDA, ROCm, Linux |
| **Languages** | Python, C/C++, Java, Javascript, Bash, MATLAB, VHDL |

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|---|---|
| **Frameworks** | TensorFlow, Pytorch, Numpy, Scipy, Matplotlib, Plotly, Pandas, Jax |
| **Tools** | Vim, Git, Docker, LaTeX |
| **Audio** | Over a decade of experience as a FOH engineer including theater productions and worship services |

## References

References available upon request